EPSR-test-ex9 One X-ray data set

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the X-ray branch of the `epsr` engine [1,2]: a data set whose weights file is an X-ray weights file (`*.XWTS.wts`, written by the `xwts` tool). X-ray data are weighted by Q-dependent atomic form factors instead of constant scattering lengths, so the weights, the self-scattering background and the EP update all take separate code paths.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `get_wts` (X-ray header and form-factor coefficients), `wxtype` (f(Q) and modified form factors), `wxcreate` (Q-dependent weights, writes `X-raywts.dat`, `X-rayf0x.dat`)
- `epsr_standalone_rev3.f`: `runepsr17` and `ptgnsq` X-ray branches (`xfile`), `interp` (data interpolated onto the Q grid)
- `epsr_standalone_rev3.f`: `inita` (`iinit 1`)

Rust (port, rust/epsr-core/src/):

- `weights.rs`: `State::get_wts`, `read_xray_weights`, `wxtype`, `wxcreate`
- `epsr.rs`: `Epsr::iterate` X-ray branches; `potential.rs`: `State::ptgnsq`
- `structure.rs`: `interp`, `State::inita`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [3] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). Data sets 1 and 2 are the neutron data of deuterated DMSO in D2O and in null-scattering H2O/D2O (ISIS SANDALS runs SLS03863, SLS03871) [4]. Data set 3 is treated as X-ray data: its weights use Waasmaier-Kirfel form-factor coefficients [5] (`f0_WaasKirf.dat`) for S, C, O, H, normalised to $\langle f^2 \rangle$ with independent-atom form factors (header `1 2 1`). Its data file `DMSOWater_xray.mint01` is a byte-identical copy of the H2O neutron data (SLS03850), so the case tests the code path, not a physical X-ray fit.

Inputs

Copy of EPSR-test with `iinit 1` and data set 3 replaced by `DMSOWater_xray.mint01` + `DMSOWater.XWTS.wts` in the `.EPSR.inp`; the weights were made with `xwts` from `DMSOWater.XWTS.dat`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Two chained passes (pass 1 with `iinit 1`, pass 2 continuing). Besides the 27 standard outputs, the X-ray files `X-raywts.dat` (1200 lines) and `X-rayf0x.dat` (600 lines) are compared; all 29 files are identical after both passes.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] D. Waasmaier and A. Kirfel, "New analytical scattering-factor functions for free atoms and ions", Acta Cryst. A51, 416-431 (1995).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.